José Ignacio Maldonado Pérez

Computer Systems Engineer — Full-Stack Development & AI Automation

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PROFESSIONAL SUMMARY

Computer Systems Engineer with 3+ years of experience in network operations, server testing and diagnostics, and teaching. I complement that foundation with full-stack development and AI automation: I have designed, deployed and currently maintain two production apps that integrate language models, third-party APIs and Google Cloud services. Looking for a development or automation role where I can take a project from idea to deployment.

TECHNICAL SKILLS

Languages	JavaScript (Node.js) · Python and Java (basic) · SQL · HTML/CSS
Backend & APIs	Node.js, Express 5, REST, OAuth 2.0, Firebase Cloud Functions, third-party APIs
Frontend	React 19, Tailwind CSS, shadcn/ui
AI & automation	Google Gemini API, Google Cloud TTS, n8n, prompt design, content pipelines
Cloud & infra	Google Cloud Run, Cloud Storage, Firebase, Docker, Linux, Git/GitHub
Data & quality	Firestore, Power BI, node and server testing, functional testing, documentation

SELECTED PROJECTS

CIENCIA COOL

Personal project · 2026

AI-powered automated video generation and publishing platform · Node.js · Google Cloud · n8n · FFmpeg · Demo / Code

- End-to-end pipeline in Node.js and Express: picks news items from RSS feeds, writes the script with Gemini, synthesises narration with Google Cloud TTS, selects stock footage and renders the video to MP4 with FFmpeg.
- Deployed on Google Cloud Run with Cloud Storage, orchestrated by self-hosted n8n on Docker; integrated OAuth 2.0 with YouTube, Instagram, TikTok and Facebook, with automatic token refresh and scheduled publishing to all four networks.
- Production hardening: API-key auth, duplicate-run lock (409/429), idempotency by job id and Cloud Run payload size limits.

NEXO BOT

Personal project · 2026

WhatsApp assistant for businesses: menu, booking and FAQs with AI · Node.js · WhatsApp Cloud API · Gemini · Firestore · Demo

- WhatsApp assistant with an interactive menu, appointment booking against real availability, FAQs and an owner notification every time a booking comes in.
- Conversation engine in Node.js 22 and Express 5 with Gemini and a local FAQ fallback: if the model fails or the quota runs out, the menu and bookings keep working.
- Webhook with HMAC-SHA256 signature validation, a swappable storage adapter (Firestore or in-memory) and the same engine across WhatsApp, web and CLI. 11 automated tests.

WORKLISH

Personal project · 2026

Professional English app with a conversational AI tutor · React · Firebase · Gemini · Demo

- Web and Android app for practising professional English with an AI tutor: exercises, corrections and voice conversation powered by Gemini.
- Frontend in React 19, Tailwind CSS and shadcn/ui; backend on Firebase Cloud Functions (Node.js 20) with Authentication and Firestore, security rules and indexes, plus a signed Android APK.

PROFESSIONAL EXPERIENCE

FOXCONN

Jun 2025 – present

Jr. Test Engineer · Electronics manufacturing, NVIDIA project

- Run tests on nodes and servers and diagnose hardware and software failures.
- Support the NVIDIA account: reproduce defects, document findings and track reported cases.

UNIVERSIDAD DE DESARROLLO NACIONAL (UDN)

May 2024 – Sep 2025

Lecturer, Engineering Department

- Taught engineering courses and developed course content and teaching materials.

MEGACABLE

Sep 2022 – Dec 2023

Jr. NOC Operations Engineer · Networks and telecommunications

- Handled service incidents, classified them by severity and followed up in line with SLAs.
- Built and analysed Power BI reports to identify operational improvement opportunities.

EDUCATION, CERTIFICATIONS & LANGUAGES

Education	B.Eng. in Computer Systems Engineering, UDN, 2018–2021. Federal professional licence 13727204.
Certifications	EF SET English Certificate B1 · Power BI Career Accelerator · Data, Data, Everywhere · Ask Questions to Make Data-Driven Decisions (last three from Coursera).
Languages	Spanish (native) · English B1, EF SET certified.